EVALUATION OF NLP SYSTEMS

Lecture 6 - Statistical Significance Testing

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- what's it
- warnings
- two tests

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NLP nowadays is very empirical

a lot of papers rely only on experiments

state-of-the-art is claimed based on comparison of systems' performance

what can go wrong?

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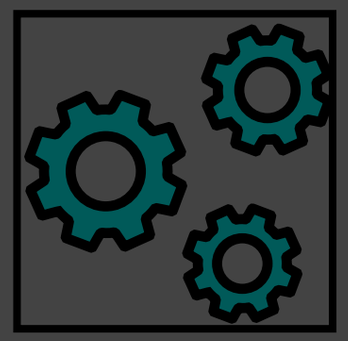
you have built an NLP system

system A

and you want to show that it's better than another system

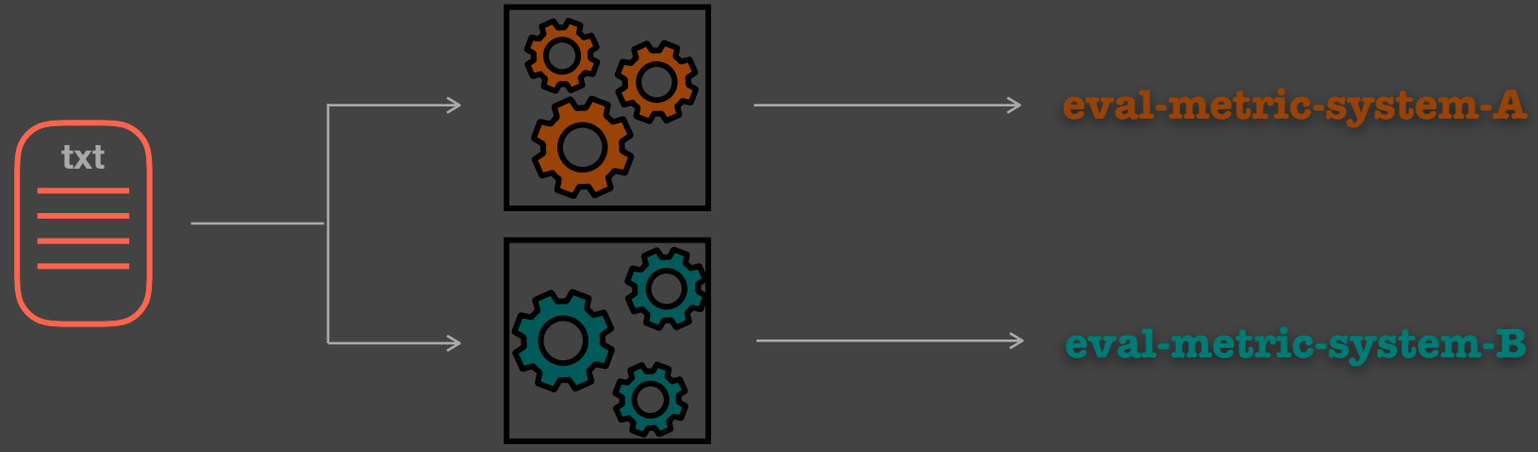


system B

what do you do?

test both systems

find a test set X



get their evaluation results

if eval-metric-system-A is better than eval-metric-system-B, your system performance is superior.

ARE YOU SURE?

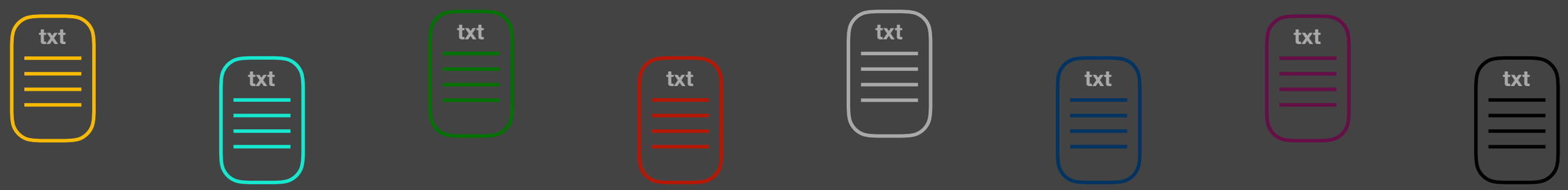
IT'S NOT THAT SIMPLE.

all you can affirm is that, in that specific test set, system A performs better than system B with respect to that specific metric

you can try other metrics - that will give more evidence

but what if it's only pure luck?

what would happen with other test sets out there?



you cannot know when they do not exist (and they usually don't)

SO WE RESORT TO STATISTICAL SIGNIFICANCE TESTS.

formal statistical inference methods to measure how likely an outcome is under given circumstances

eval-metric-system-A

→ let's call it $\mathbb{M}_A$

eval-metric-system-B

→ let's call it $\mathbb{M}_B$

for example, M can be the F1 Score

'advantage' of system A over system B wrt. any arbitrary test set Y

$$\mathbb{M}_A(Y) - \mathbb{M}_B(Y) = \delta$$

called 'test statistic', has its own (usually unknown) probabiliy distribution

if system A is better than system B, this difference is a positive number

'advantage' of system A over system B wrt. your test set X

$$\mathbb{M}_A(X) - \mathbb{M}_B(X) = \delta_{obs}$$

the specific value you observed in your test set

if system A was not better than system B, what's the probability of observing δ_{obs}?

meaning: if system B was better or if A and B were equally good, how likely would it be to get the difference you observed with test data X?

we formulate a null hypothesis $H_0: \delta \leq 0$ and estimate a p-value

and an alternative hypothesis $H_1: \delta > 0$

p-value $= P(\delta \geq \delta_{obs} \mid H_0$ **is true**$)$

a very small p-value is desired

if p-value < α, you have evidence to reject the null hypothesis

alpha has to be defined BEFORE the test!

Type I

reject the null hypothesis when it is, in reality, true

Two types of errors

Type II

fail to reject the null hypothesis when it is, in reality, false

---

we never 'accept' a hypothesis!

we gather evidence to either **reject** or **fail to reject** the null hypothesis

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there are many kinds of tests: parametric, nonparametric, paired, with or without sampling…

- Paired Z-test
- Paired Student's t-test
- ANOVA
- Pearson's r-test
- Sign test
- McNemar'stest
- Wilcoxon signed-rank test
- Fisher–Pitman permutation test
- Paired bootstrap test

each has specific assumptions that must be met, otherwise the results are not reliable!

more details are covered
in this week's reading

PAUSE THIS VIDEO AND WATCH THIS TALK

HTTPS://VIMEO.COM/285803636

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DON'T TRUST THESE TESTS TOO MUCH EITHER!
USE WITH CAUTION!

⚠ WARNINGS ⚠

Statistical significance tests commonly used in NLP setups share the assumption that the data samples are independent and identically distributed (iid).

This assumption is rarely true in NLP setups, which have dependent observations (texts in a corpus).

The unique structured nature of natural language data is reflected in specialized evaluation measures.

We still lack specialized statistical significance tests. So far, we import them from other areas. Statisticians needed.

It is not uncommon to find papers that report significance but use the wrong test. Or that do not report it at all.

While cross-validation is a desired practice, it is challenging to report statistical significance when it is employed.

The word significant is usually misused to mean 'important' or 'substantial' in the literature. We should be more careful to retain its statistical sense.

Dror et al. (2018)

WSJ FALLACY: the false belief that performance on the test section of the Wall Street Journal (WSJ) part of the English Penn treebank is representative for performance on other texts in English; more generally, the belief that our samples are always representative (sample bias)

sometimes researchers violate test assumptions and basic statistical methodology, e.g., when there is no *one* established metric, when we can't run our models on full-length sentences, or when data is biased

p-value cut-offs in other areas (e.g. physics) is probably not suitable for NLP, but we use them

BEWARE OF P-VALUES!

they are sensitive to sample size, covariates, multiple metrics

cherry-picking datasets and metrics increases the *a priori* chance of establishing significance

system A and system B may perform equally well as measured by some metric, but contribute very differently to downstream tasks

good practice: predict significance across datasets and report results across available metrics

most NLP metric for comparing system outputs can be shown to be non-normally distributed, so many significance tests are not suitable

Søgaard, Johannsen, Plank, Hovy & Martinez (2014)

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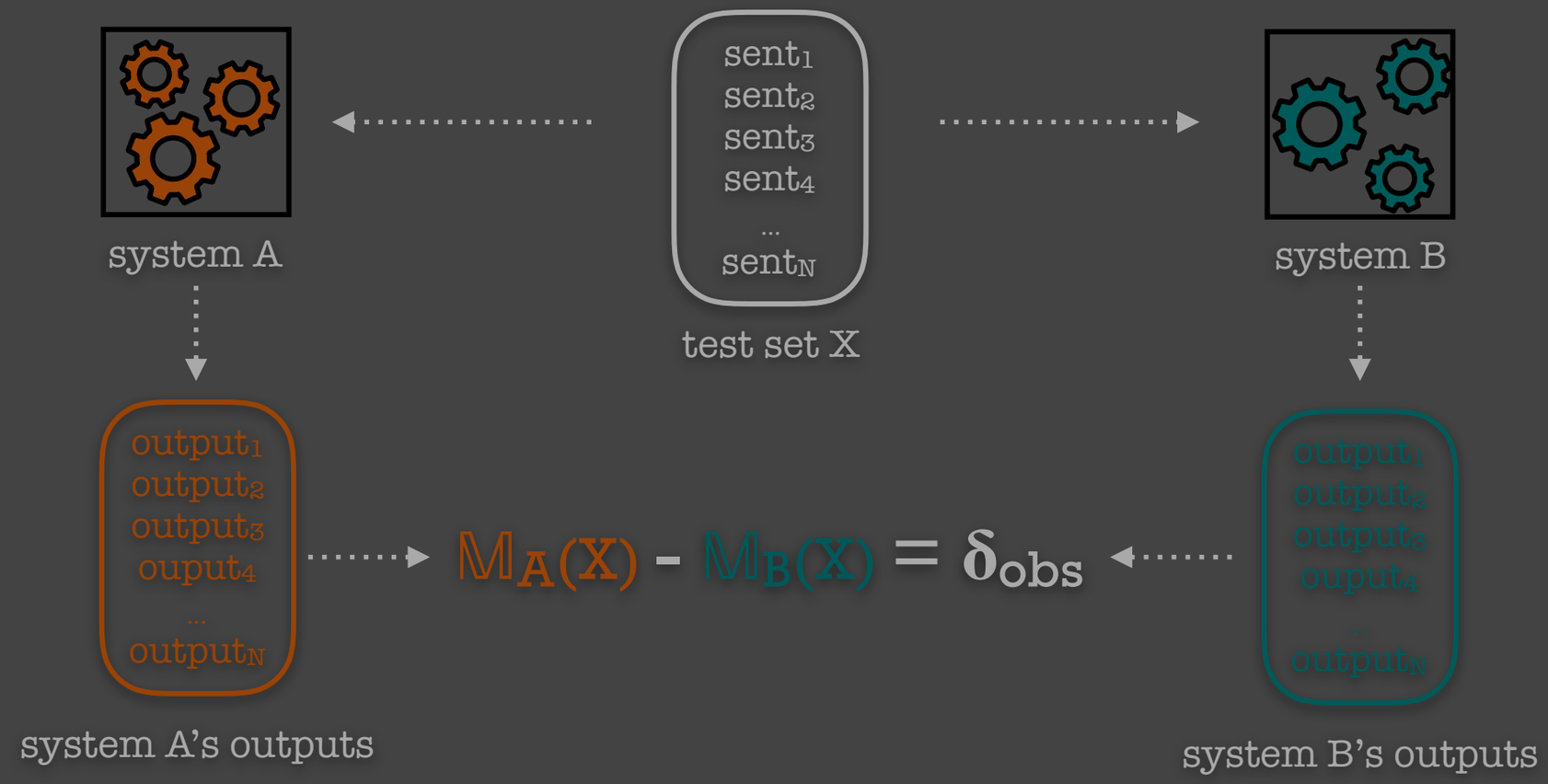
PAIRED APPROXIMATE RANDOMIZATION TEST



initialize:

```
i = 0
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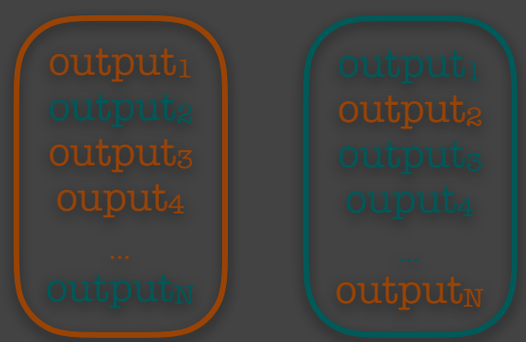
repeat T times:



randomly swap the outputs

pretend these are system A and B outputs

$$\mathbb{M}_A(X) - \mathbb{M}_B(X) = \delta_i$$

collect a new estimation of δ

if $\delta_i > \delta_{obs}$: i+=1

check if δ_i is greater than δ_{obs} and increment counter

finally:

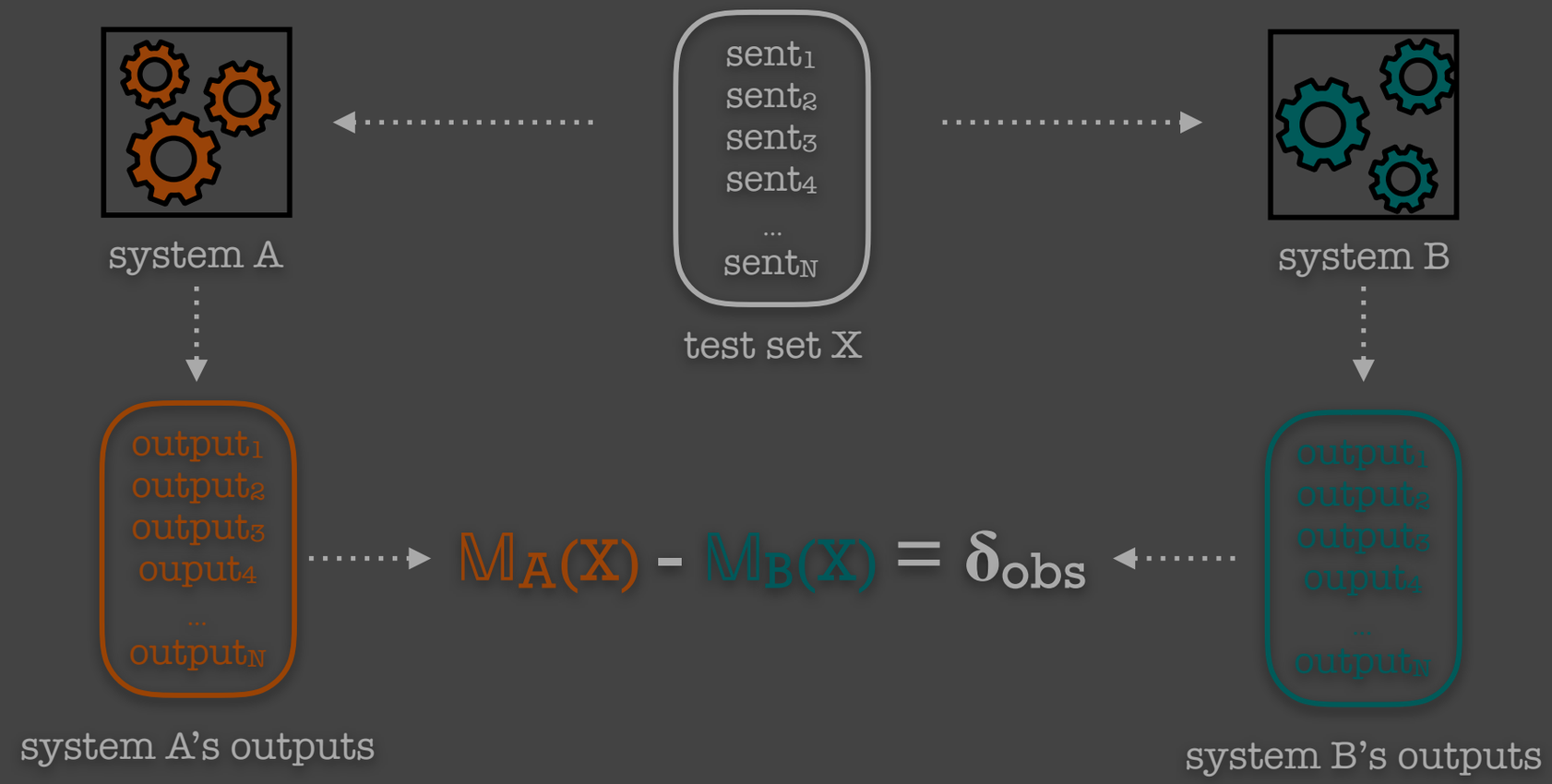
$$p \approx (i+1)/(T+1)$$

estimate the p-value

the estimation gets better as T goes to infinity

Noreen (1989)

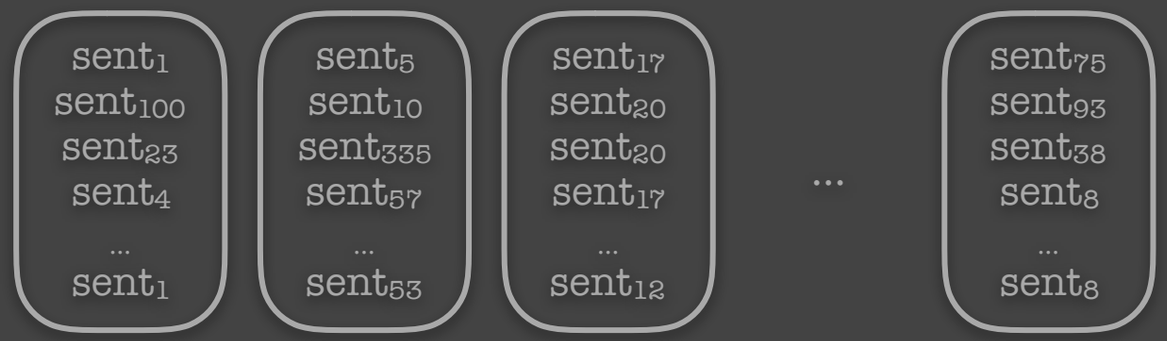
PAIRED BOOTSTRAP TEST



initialize:

$s = 0$

pretend that X is the whole population and create b pseudo test sets of size N by sampling from X with replacement



for each created test set Xi:

get metric and increment s if it's greater than observed metric

$$M_A(X_i) - M_B(X_i) = \delta_i$$

$$\text{if } \delta_i > 2\delta_{obs} : s{+}{=}1$$

finally:

$p \approx s/b$ estimate the p-value

b must be large enough

Berg-Kirkpatrick, Burkett & Klein (2012)
Efron and Tibshirani (1993)

See you next week!

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